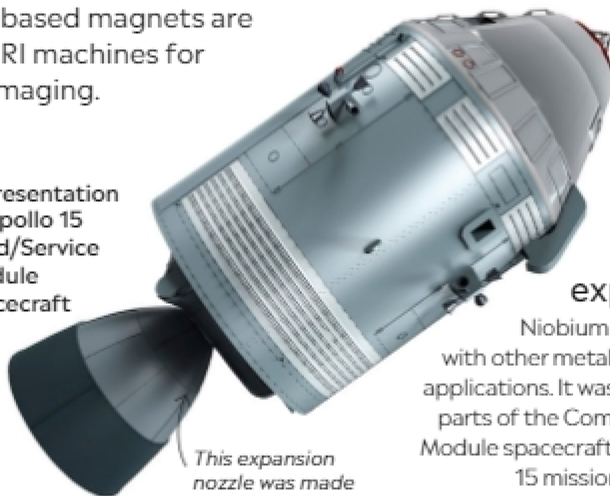
Niobium

Rods of pure niobium refined in a laboratory

41 Nb Sourced mainly from the mineral columbite, the transition metal niobium has many uses: from pacemaker cases for the human heart to commemorative coins, jet engines, and rocket parts. Niobium-based magnets are used in MRI machines for medical imaging.



Artist's representation of the Apollo 15 Command/Service Module (CSM) spacecraft

This expansion nozzle was made of a niobium alloy.

Space exploration

Niobium is often mixed with other metals for specialist applications. It was used to make parts of the Command/Service Module spacecraft for the Apollo 15 mission to the moon.

Technetium

43 Tc Originally thought not to occur in nature, it was later discovered that tiny amounts of technetium do exist as the result of the natural radioactive decay of uranium and thorium. It was not easy to find the element. Mendeleev predicted its existence as a missing element on his original periodic table. There followed many false claims of discovery, before Italian Carlo Perrier and Italian American Emilio Segrè finally isolated two of its isotopes in 1937. Its name comes from a Greek word meaning "artificial" because it was the first artificially produced element. Today, it is used mainly in medical imaging.

Ruthenium

44 Ru This transition metal is mostly obtained as a by-product when other metals, such as nickel and copper, are purified. Ruthenium is often combined with platinum to harden electrical components. The element can also help speed up the industrial production of ammonia, a key ingredient in fertilizers.

Bright, silvery metal

Pure ruthenium refined in a laboratory

Rhodium

45 Rh This rare and hard metal is mainly obtained as a by-product during the refining of nickel and copper. One of rhodium's main applications is in making catalytic converters—devices that convert harmful exhaust gases into less dangerous ones—for cars. It is also used in many alloys to increase their resistance to corrosion.

Pure rhodium pellet refined in a laboratory

Rhodium-coated mirror makes the reflection of an object precise and sharp.

Special mirrors

Rhodium's ability to be polished to a bright shine means that it can be used in industrial and specialized mirrors, such as the mouth mirrors used by dentists.